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| United States Patent | 12410648           |
| Kind Code            | B2                 |
| Date of Patent       | September 09, 2025 |
| Inventor(s)          | Kellum; Wilbur J.  |

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### Window balance shoes for a pivotable window

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#### Abstract

A balance shoe for a block and tackle window balance system includes an enlarged head portion housing a locking system configured to receive at least a portion of a pivot bar and releasably engage a jamb track. An elongate tail portion configured to couple at least partially within a U-shaped channel of the window balance system. A front face, the front face of the elongate tail portion being adjacent to a base wall of the U-shaped channel when the elongate tail portion is coupled therein. The front face including an elongate channel configured to allow passage of the pivot bar from the elongate tail portion towards the locking system. The balance shoe also including at least one protrusion extending from the front face of the elongate tail portion and disposed at least partially within the elongate channel.

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| <b>Family ID:</b> | <b>65721397</b>                          |
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| <b>Appl. No.:</b> | <b>18/795355</b>                         |
| <b>Filed:</b>     | <b>August 06, 2024</b>                   |

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20250027351 A1          | Jan. 23, 2025           |

#### Related U.S. Application Data

continuation parent-doc US 17451792 20211021 US 12091895 child-doc US 18795355  
continuation parent-doc US 16136650 20180920 US 11193318 20211207 child-doc US 17451792  
us-provisional-application US 62561580 20170921

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## Publication Classification

**Int. Cl.:** E05D15/22 (20060101); E05C17/60 (20060101); E05D13/00 (20060101)

**U.S. Cl.:**

**CPC** E05D15/22 (20130101); E05D13/04 (20130101); E05D13/1207 (20130101);  
E05Y2201/67 (20130101); E05Y2900/148 (20130101)

## Field of Classification Search

**USPC:** None

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/451,792, now U.S. Pat. No. 12,091,895, filed Oct. 21, 2021, which is a continuation of U.S. patent application Ser. No. 16/136,650, now U.S. Pat. No. 11,193,318, filed Sep. 20, 2018, which claims priority to and the benefit of U.S. Provisional Patent Application No. 62/561,580, filed on Sep. 21, 2017, the entire disclosures of which are incorporated herein by reference in their entireties.

## INTRODUCTION

(1) Pivotal double hung windows can include two window sashes disposed in tracks located in a window frame to allow vertical sliding movement of the sashes. Pivot bars can be provided to allow rotational movement of the window sashes about the pivot bars to facilitate cleaning and/or removal of the sash. To control vertical movement, window balances are used so that the window sashes remain in a position in which they are placed. Balance shoes are also used to guide the rotational movement of the window sashes with respect to the window frame, as well as lock the window sash in position when pivoted.

(2) Various types of balance systems are known and are used to counterbalance the weight of the window sash. For example, block and tackle systems include a system of pulleys and an extension spring mounted within a rigid channel, and are relatively compact in size and easy to install.

## SUMMARY

(3) In an aspect, the technology relates to a balance shoe for a block and tackle window balance system, the balance shoe including: an enlarged head portion housing a locking system configured to receive at least a portion of a pivot bar and releasably engage a jamb track; an elongate tail portion configured to couple at least partially within a U-shaped channel of the window balance system; a front face, wherein the front face of the elongate tail portion is adjacent to a base wall of the U-shaped channel when the elongate tail portion is coupled therein, and wherein the front face includes an elongate channel configured to allow passage of the pivot bar from the elongate tail portion towards the locking system; and at least one protrusion extending from the front face of the elongate tail portion and disposed at least partially within the elongate channel.

(4) In an example, the at least one protrusion is configured to engage with the base wall of the U-shaped channel. In another example, the at least one protrusion engages with the base wall in a resilient connection. In still another example, the at least one protrusion is configured to restrict a pullout force of the elongate tail portion from the U-shaped channel. In yet another example, the at least one protrusion is configured to transfer a load between the elongate tail portion and the U-shaped channel in shear. In an example, a rear face is opposite of the front face, and the at least one protrusion includes a face surface that is substantially parallel with the rear face.

(5) In another example, the at least one protrusion includes a curved top wall. In still another example, at least one connecting device extends from the elongate tail portion and is configured to engage with a sidewall of the U-shaped channel. In yet another example, the elongate tail portion defines a longitudinal axis, and the at least one connecting device includes an arm extending along the longitudinal axis. In an example, a projection extends from the arm, and the projection tapers in a direction that is outward from the arm and away from the front face.

(6) In another aspect, a block and tackle window balance system including: a U-shaped channel including a base wall and two opposing sidewalls housing at least partially a block and tackle balance assembly, wherein the U-shaped channel includes a first end having a fastener extending between the two sidewalls and at least one opening defined within the base wall; and a balance shoe coupled to the fastener, wherein the balance shoe includes: an enlarged head portion extending from the first end of the U-shaped channel; a locking system housed within the enlarged head portion configured to receive at least a portion of a pivot bar and releasably engage a jamb track; an elongate tail portion received at least partially within the U-shaped channel; a front face adjacent to the base wall of the U-shaped channel, wherein the front face includes an elongate channel

configured to allow passage of the pivot bar from the elongate tail portion towards the locking system; and at least one protrusion extending from the front face of the elongate tail portion and disposed at least partially within the elongate channel, wherein the at least one protrusion engages with the at least one opening.

(7) In an example, the base wall includes a ramped portion at the first end of the U-shaped channel. In another example, the ramped portion corresponds in size and shape to the elongate channel and is configured to allow passage of the pivot bar towards the locking system. In still another example, the at least one opening corresponds in size and shape to the at least one protrusion. In yet another example, the balance shoe further includes two connecting devices, each extending from opposite sides of the elongate tail portion and configured to engage with the two sidewalls of the U-shaped channel. In an example, an aperture is defined in each sidewall of the U-shaped channel at the first end, and the aperture is configured to receive at least a portion of the corresponding connecting device. In another example, a dimple is formed in the sidewall of the U-shaped channel proximate the aperture.

(8) In another aspect, the technology relates to a method of assembling a block and tackle window balance system, the method including: engaging a balance shoe with a fastener extending across a U-shaped channel in a first orientation, wherein the U-shaped channel includes a base wall and two opposing sidewalls; pivoting the balance shoe into a different second orientation relative to the U-shaped channel, wherein in the second orientation an elongated tail portion of the balance shoe is disposed at least partially within the U-shaped channel and an enlarged head portion of the balance shoe extends from the U-shaped channel; and substantially simultaneously with pivoting the balance shoe, engaging at least one protrusion of the balance shoe with at least one corresponding opening defined in the base wall of the U-shaped channel, wherein the at least protrusion extends from a front face of the balance shoe and at least partially within an elongate channel of the balance shoe, and wherein the elongate channel is positioned adjacent to the base wall of the U-shaped channel in the second orientation and is configured to allow passage of a pivot bar.

(9) In an example, the method further includes engaging at least one connecting device of the balance shoe with a sidewall of the U-shaped channel. In another example, engaging the at least one connecting device includes sliding at least a portion of the connecting device through a dimple formed in the sidewall of the U-shaped channel.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) There are shown in the drawings, examples which are presently preferred, it being understood, however, that the technology is not limited to the precise arrangements and instrumentalities shown.

(2) FIG. 1 is a perspective view of a pivotable double hung window assembly.

(3) FIG. 2 is a rear view of an exemplary window balance system.

(4) FIG. 3A is a perspective view of an exemplary window balance shoe of the window balance system shown in FIG. 2.

(5) FIG. 3B is a perspective view of the window balance shoe shown in FIG. 3A coupled to a U-shaped channel of the window balance system shown in FIG. 2 in a locked configuration.

(6) FIG. 3C is a perspective view of the window balance shoe shown in FIG. 3A coupled to a U-shaped channel of the window balance system shown in FIG. 2 in an unlocked configuration

(7) FIGS. 4A and 4B are schematic views of loading forces that act on the window balance system shown in FIG. 2.

(8) FIG. 5 is a perspective view of another window balance shoe.

(9) FIG. 6 is a side view of an exemplary protrusion.

(10) FIG. 7 is a perspective view of another window balance shoe.

(11) FIG. 8 is a flowchart illustrating a method of assembling a block and tackle window balance system.

#### DETAILED DESCRIPTION

(12) The examples of a balance shoe for an inverted block and tackle window balance system described herein provide a more robust connection between the shoe and the U-shaped channel. Accordingly, performance and efficiency of the installation and operation of the window balance system is increased. Additionally, heavier window sashes may be supported within the hung window assembly. In aspects, the balance shoe includes a front facing elongated channel that allows passage of a pivot bar to the locking system of the shoe. One or more protrusions are disposed within the elongated channel that are configured to engage with the U-shaped channel of the block and tackle balance system. The protrusion is configured to increase the shear strength of the shoe and U-shaped channel connection. Additionally, the protrusion can increase the pullout force required to disengage the shoe from the U-shaped channel. In other aspects, the balance shoe includes a connection device that is configured to engage with the sidewalls of the U-shaped channel. The connection devices have a flexible arm with a projection extending therefrom. The length of the flexible arm is reduced compared to known balance shoes so as to increase the strength of the connection device and U-shaped channel connection. Additionally, the projection includes a tapered section that increases the wear resistance of the connection device.

(13) FIG. 1 is a perspective view of a pivotable double hung window assembly **100** for which a window balance shoe as described herein may be used. The pivotable double hung window assembly **100** includes a window frame **102**, a pivotable lower window sash **104**, a pivotable upper window sash **106**, and a window jamb **108**. The pivotable lower window sash **104** and the pivotable upper window sash **106** slide vertically in a jamb track **110** within the window jamb **108**, while also being able to pivot about a pivot bar **112**. Each window sash **104**, **106** includes a top sash rail **114**, a base sash rail **116**, and a pair of vertical stiles **118**. In other examples, the window assembly **100** may be a single hung window assembly in which only the lower window sash **104** is pivotable and slidable.

(14) A window balance system **200** is mounted within the jamb track **110** and provides a counter balance force to the window sashes **104**, **106**. Additionally, the window balance system **200** guides the pivoting movement of the window sashes **104**, **106** and locks in position within the jamb track **110** during the pivoting and/or removal of the window sashes **104**, **106**. In the example, the window balance system **200** is a block and tackle window balance system, although other balance systems (e.g., constant force balances) may be used as required or desired. The window balance system **200** is described in reference to FIG. 2 below.

(15) FIG. 2 is a rear view of the exemplary window balance system **200**. The window balance system **200** includes an inverted window balance **202** that is used for balancing the weight of the window sash within the window frame, and a window balance shoe **204** for guiding the rotation of the window sash about a pivot bar as described above. In the example, the inverted window balance **202** is a block and tackle type window balance and includes an extension spring **206** connected to a system of pulleys **208** housed within a rigid U-shaped channel **210**. A cord **212** connects the system of pulleys **208** to a jamb mounting attachment **214**, such as a cord terminal or hook, at a top end **215** of the U-shaped channel **210**. Opposite the jamb mounting attachment **214**, a fastener **216** (e.g., a rivet) extends across the U-shaped channel **210** at a bottom end **217** of the U-shaped channel **210**. The balance shoe **204** is coupled (e.g., resiliently secured) to the inverted window balance **202** at the bottom end **217** of the U-shaped channel **210**.

(16) The balance shoe **204** includes a substantially T-shaped body **218** with an elongate tail portion **220** that is configured to be at least partially received within the U-shaped channel **210** and couple to the fastener **216**, and an enlarged head portion **222** that extends from the bottom end **217** of the U-shaped channel **210**. The elongate tail portion **220** includes one or more connecting devices **224**

that engage with the U-shaped channel **210** and enable the balance shoe **204** to at least be partially secured within the U-shaped channel **210**. The enlarged head portion **222** houses a locking system **225** that is configured to receive at least a portion of the pivot bar of the window sash and releasably engage the jamb track. The locking system **225** includes a rotatable cam **226** and a locking device **228**. The locking device **228** surrounds the cam **226** and includes a pair of opposing ends **230** connected by a spring member **232**. The cam **226** is configured to receive the pivot bar of the pivotable window sash such that when the sash is tilted open, the pivot bar rotates, thereby rotating the cam **226** and forcing the opposing ends **230** of the locking device **228** outward from the enlarged head portion **222** to engage the jamb track of the window frame and to lock the balance shoe **204** in location.

(17) Additional examples of T-shaped balance shoes that may be used with the inverted window balance, methods of assembly of inverted window balances, and methods of installation thereof are described further in U.S. Pat. No. 6,679,000, filed Jan. 11, 2002, and entitled “SNAP LOCK BALANCE SHOE AND SYSTEM FOR A PIVOTABLE WINDOW,” the disclosure of which is hereby incorporated by reference herein in its entirety.

(18) FIG. 2 illustrates the rear view of the window balance system **200**, which is the side that faces the jamb track when mounted within the window jamb. As such, the U-shaped channel **210** at least partially covers the block and tackle assembly (e.g., the spring **206**, the pulleys **208**, and the cord **212**) and restricts access thereto. Additionally, the U-shaped channel **210** restricts or prevents dirt and debris from accumulating on the block and tackle assembly. In this orientation, however, the bottom end **217** of the U-shaped channel **210** may make sash installation more difficult due to the proximity of the bottom end **217** to a keyhole opening **273** (shown in FIGS. 3B and 3C) in the cam **226**. As such, the U-shaped channel **210** and the balance shoe **204** include features that enable the pivot bar to more easily pass to the cam **226** and make sash installation more efficient.

(19) FIG. 3A is a perspective view of the window balance shoe **204** of the window balance system **200** (shown in FIG. 2). In FIG. 3A the locking system **225** (also shown in FIG. 2) is not illustrated for clarity. In the example, the shoe body **218** has a front face **234** that is configured to allow passage of the pivot bar to the locking system during window sash installation and an opposite back face **236** that is configured to slide against the jamb track wall. As such, the front face **234** is configured to mount into and adjacent to the U-shaped channel **210** (shown in FIG. 3B), and the back face **236** is substantially planar with the U-shaped channel **210** so that the balance shoe **204** can slide up and down in the window jamb during use. The balance shoe **204** enables the window sash, via the pivot bar, to be coupled to the window balance and facilitate the vertical sliding movement and the pivoting movement of the window sash within the window jamb.

(20) A connection pocket **238** is defined in the back face **236** of the elongate tail portion **220** and towards a top end **237** of the balance shoe **204**, which is opposite the enlarged head portion **222**. Thus, the top end **237** of the balance shoe **204** defines a hook **239** that is configured to secure to the fastener **216** of the U-shaped channel **210** (both shown in FIG. 3B). The hook **239**, when engaged with the fastener **216** (shown in FIG. 2), enables the window load supported by the balance shoe **204** to be transferred (e.g., via shear force resistance) to the U-shaped channel and the block and tackle balance components. As such, when the balance shoe **204** is connected to the U-shaped channel and installed in the window jamb, the weight of the window is supported by the balance shoe **204** so that the window sash can move along a longitudinal axis **241**. This longitudinal axis **241** is substantially parallel to the jamb channel and is also substantially parallel with a longitudinal axis of the balance shoe **204**. In the example, the connection pocket **238** extends from the back face **236** toward the front face **234** and is sized and shaped to receive the fastener **216** (shown in FIG. 2). For example, the connection pocket **238** is substantially orthogonal to the back face **236** and the longitudinal axis **241**. In alternative examples, the connection pocket **238** may be angled or curved so as to receive the fastener, or may be a through-hole entirely defined by the elongate tail portion **220** such that the fastener extends therethrough.

(21) The elongate tail portion **220** is sized and shaped to be received and secured within the U-shaped channel **210** (shown in FIG. 3B). In addition to the connection pocket **238**, the elongate tail portion **220** includes additional features that enable the balance shoe **204** to be received within the U-shaped channel **210** and prevent the shoe **204** from rotating about the fastener and out of the U-shaped channel, which is undesirable. In the example, the elongate tail portion **220** includes two opposing sidewalls **240** that extend between the front face **234** and the back face **236**, and along the longitudinal axis **241**. Each sidewall **240** includes the connecting device **224**, which is configured to be resiliently secured to the U-shaped channel and prevent the elongate tail portion **220** from disengaging from the U-shaped channel (e.g., pulling out of the U-shaped channel and/or from rotating out of the U-shaped channel).

(22) The connecting devices **224** include a resilient, flexible arm **242** extending along and substantially parallel to the longitudinal axis **241** of the elongate tail portion **220**. The arm **242** includes an engagement projection **244**, such as a tab, located at the free end of the arm **242** and extending outwards from the elongate tail portion **220**. The projection **244** is shaped and sized to engage with a corresponding aperture **266** (shown in FIG. 3B) defined in the U-shaped channel and lock the balance shoe **204** to the U-shaped channel. In the example, the projection **244** tapers in a direction that is outwards from the arm **242** (e.g., substantially perpendicular to the longitudinal axis **241**) and away from the front face **234** towards the back face **236**. That is, the height of the projection **244** is greater at the back face **236** than the front face **234**. This taper of the engagement projection **244** provides added material to the side of the projection **244** that slides against the U-shaped channel to reduce or eliminate wear on the projection **244** during shoe installation.

(23) Each resilient arm **242** is disposed substantially parallel to an adjacent sidewall **240**, but spaced therefrom, and is configured to deflect towards the longitudinal axis **241**. As such, the arm **242** is connected to and extends from the sidewall **240** at a line of flexure **245**, and may be at least partially skew to the longitudinal axis **241**. Because of the flexure of the arm **242**, when the balance shoe **204** is inserted within the U-shaped channel, the U-shaped channel forces the arm **242** to deflect until the engagement projection **244** engages with the U-shaped channel. More specifically, the engagement projection **244** is configured to engage, for example, via a resilient-fit connection, with a corresponding aperture **266** of the U-shaped channel **210** (shown in FIG. 3B). Additionally, the arm **242** has a length **246** that is sized so as to reduce excessive flexure and wear to the connecting device **224**. Excessive flexure may prevent engagement between the engagement projection **244** and the U-shaped channel. Furthermore, reducing the length **246** of the arm **242** enables engagement with the U-shaped channel to be strengthened by providing a greater retention force generated by the flex of the arm **242**. In the example, the connecting devices **224** may be used in concert with the hook **239**/connection pocket **238** to at least partially secure the balance shoe **204** to the inverted window balance **202** (shown in FIG. 2). In other examples, the connecting device **224** may be used without the hook **239**/connection pocket **238** to couple the balance shoe **204** to the inverted window balance.

(24) An elongate channel **248** is defined within the front face **234** of the balance shoe **204** and within elongate tail portion **220**. The elongate channel **248** extends from approximately the top end **237** of the elongate tail portion **220** towards a cam opening **250** defined in the enlarged head portion **222**. The cam opening **250** is sized and shaped to house the cam **226** (shown in FIG. 2) and enable the cam to rotate therein to extend and retract the ends **230** (also shown in FIG. 2) and lock and unlock the balance shoe **204** within the window jamb. The elongate channel **248** is recessed within the front face **234** and allows passage of the pivot bar from the elongate tail portion **220** towards the cam opening **250**. As such, the pivot bar may be easily inserted into the cam during window sash installation (e.g., without the need to rack the window). The elongate channel **248** is ramped or pitched and extends from the top end **237** of the elongate tail portion **220** to a lead-in lip **251** proximate the cam opening **250**, to facilitate guiding the pivot bar towards the enlarged head portion **222** and into the cam keyhole opening **273** (shown in FIGS. 3B and 3C). That is, the

elongate channel **248** extends deeper in depth D.sub.1 within the front face **234** at the lip **251** than at the top end **237** of the elongate tail portion **220** at a depth D.sub.2.

(25) The elongate tail portion **220** also includes a protrusion **252** that extends from the front face **234** of the elongate tail portion **220** and is disposed at least partially within the elongate channel **248**. The protrusion **252** is sized and shaped to be received within an opening **268** (shown in FIG. 3B) defined within the U-shaped channel **210** so that the balance shoe **204** is engaged with the U-shaped channel at yet another location. In the example, the protrusion **252** is substantially rectangular or square shaped. In other examples, the protrusion **252** may have any other shape that enables the balance shoe **204** to function as described herein, such as, circular, rectangular, mushroom-shaped (see FIG. 6), triangular, linear, and the like. In the example, the protrusion **252** acts in concert with either or both of the hook **239**/connection pocket **238** and the connecting devices **224** to secure the balance shoe **204** to the inverted window balance. In other examples, the protrusion **252** may be the only connection element to the U-shaped channel. In yet other examples, the protrusion **252** may be used in concert with only the hook **239**/connection pocket **238** to secure the balance shoe **204** to the U-shaped channel.

(26) The protrusion **252** when engaged with the U-shape channel, enables the window load supported by the balance shoe **204** to be transferred (e.g., via shear force resistance) to the U-shaped channel and the block and tackle balance components. Additionally or alternatively, the protrusion **252** may be configured to engage with the U-shaped channel and prevent the elongate tail portion **220** from disengaging from the U-shaped channel (e.g., pulling out of the U-shaped channel and/or from rotating out of the U-shaped channel). In an aspect, the protrusion may be resiliently secured within the U-shaped channel and reduce or eliminate the likelihood of disconnection (e.g., a pullout force) when the window balance is transported and/or installed.

(27) In the example, the protrusion **252** has a top wall **254**, a bottom wall **256**, two sidewalls **258**, and a face surface **260** and is disposed at least partially within the elongate channel **248**. The top wall **254** and/or the bottom wall **256** may be curved. In other examples, the walls **254**, **256** may be linear and either substantially orthogonal or angled relative to the sidewalls **258**. The face surface **260** may be sloped relative to the elongate channel **248**. As such, the top wall **254** of the protrusion **252** has a smaller height than the bottom wall **256** of the protrusion **252**. Additionally, the face surface **260** is disposed below the plane of the front face **234** formed by the sidewalls **240**. This enables the pivot bar to more easily pass over the protrusion **252** when the window sash is being dropped into the balance shoe **204** and the protrusion **252** does not interfere with the operation of the hung window assembly.

(28) In some examples, the face surface **260** may be parallel to the tapered slope of the elongate channel **248**. In other examples, the face surface **260** may be substantially flat, for example, the face surface **260** may be substantially parallel with the back face **236** of the balance shoe **204**. In yet other examples, the face surface **260** may be curved or rounded. In still other examples, one or more of the walls **254-258** of the protrusion **252** may include a lip so that the protrusion **252** can more securely engage with the U-shaped channel. For example, the lip may facilitate a resilient connection between the balance shoe **204** and the U-shaped channel.

(29) As illustrated in FIG. 3A, the protrusion **252** is positioned on the elongate tail portion **220** such that it is aligned with the engagement projections **244** and is offset from the sidewalls **240**. In other examples, the protrusion **252** may be positioned at any other location on the elongate tail portion **220** as required or desired. For example, the protrusion **252** may be positioned more proximate the top end **237** of the elongate tail portion **220**. In another example, the protrusion **252** may be positioned more towards or adjacent to the sidewalls **240** of the balance shoe **204**. In yet another example, the protrusion **252** may be divided into two parts, with each part adjacent to opposite sidewalls **240**.

(30) FIG. 3B is a perspective view of the window balance shoe **204** coupled to the U-shaped channel **210** and in a locked configuration. FIG. 3C is a perspective view of the window balance

shoe **204** coupled to the U-shaped channel **210** and in an unlocked configuration. Referring concurrently to FIGS. **3B** and **3C**, the U-shaped channel **210** includes a base wall **262** and two sidewalls **264** extending therefrom. At the bottom end **217** of the U-shaped channel **210**, the fastener **216** extends between the two sidewalls **264**. Additionally or alternatively, the U-shaped channel **210** includes at least one aperture **266** defined in each sidewall **264** that is sized and shaped to receive and engage the projection **244** of the connecting device **224**. The U-shaped channel **210** also includes an opening **268** defined in the base wall **262** that is sized and shaped to receive and engage with the protrusion **252**. When the balance shoe **204** is coupled to the U-shaped channel **210**, the front face **234** of the balance shoe **204** is adjacent to the base wall **262** and the protrusion **252** is disposed in the pivot bar travel path during operation of the hung window assembly.

(31) To install and secure the balance shoe **204** within the U-shaped channel **210**, the elongate tail portion **220** is advanced at an angle into the U-shaped channel **210** so that the hook **239** (shown in FIG. **3A**) engages with the fastener **216**. That is, the fastener **216** is received within the connection pocket **238** (shown in FIG. **3A**). The balance shoe **204** is then rotated about the fastener **216** so that the front face **234** is positioned within the U-shaped channel **210** against an interior surface of the base wall **262** and between the two sidewalls **264** so as to be aligned with the balance shoe **204** along the longitudinal axis **241**. As the balance shoe **204** is rotated, the connecting devices **224** engage with the corresponding apertures **266** within the sidewalls **264** of the U-shaped channel **210**. This assembly sequence is depicted, for example, in U.S. Pat. No. 6,679,000 at FIGS. **6A-6D**, the disclosure of which is hereby incorporated by reference herein. In this example, however, one or more dimples **270** may be formed on each sidewall **264** of the U-shaped channel **210** proximate the apertures **266**. These dimples **270** extend outward from the sidewalls **264** so as to facilitate deflection of the connecting device arms while the engagement projection **244** slides into the aperture **266**. As such, wear on the connecting devices **224** is reduced during balance shoe **204** assembly. In some examples, the engagement projection **244** may be received within the aperture **266** such that the balance shoe **204** and the U-shaped channel **210** are engaged in a resilient connection.

(32) Additionally, as the balance shoe **204** is rotated into the U-shaped channel **210**, the protrusion **252** is received within and engages with the opening **268** located on the base wall **262** of the U-shaped channel **210**. In some examples, the protrusion **252** may be received within the opening **268** such that the balance shoe **204** and the U-shaped channel **210** are engaged in a resilient connection. In the example, the base wall **262** of the U-shaped channel **210** may include a ramped portion **272** that is disposed at the bottom end **217**. The ramped portion **272** tapers inward towards the sidewalls **264** so as to allow passage of the pivot bar into the balance shoe **204** during window sash installation. In the example, the ramped portion **272** may correspond in size and shape to the elongate channel **248** of the elongate tail portion **220**. This allows the ramped portion **272** in the U-shaped channel **210** to be flush or substantially flush with the lip **251** of the balance shoe **204**, thus, enabling insertion of the pivot bar to the locking system while reducing potential interference. By forming a grooved ramp in both the balance shoe **204** and the U-shaped channel **210**, wider width window sashes may be used with the window balance systems as the bottom end **217** of the U-shaped channel **210** does not block the drop-in of the pivot bars.

(33) In the example, at least a portion of the front surface **260** of the protrusion **252** extends above the U-shaped channel **210** and may be shaped and sized to direct the pivot bar up and over, or around, the protrusion **252** so that the pivot bar does not catch on the protrusion **252** as it is inserted into the cam. In other examples, the walls **254-258** of the protrusion **252** may terminate before the outer surface of the U-shaped channel **210** so that the pivot bar does not catch on the protrusion **252**. In the example, the balance shoe **204** is coupled to the U-shaped channel **210** via the hook **239** (shown in FIG. **3A**), the connecting devices **224**, and the protrusion **252**. In other examples, only the protrusion **252** and hook **239**/connection pocket **238** are used to secure the balance shoe **204** within the U-shaped channel **210**, while in yet other examples only the protrusion **252** may be



utilized.

(34) In the example, FIG. 3B illustrates the cam **226** being in a locked position such that a keyhole opening **273** is aligned with the ramped portion **272** and the elongate channel **248** (shown in FIG. 3A) so as to receive the pivot bar of the window sash. In the locked position, the ends **230** extend out of the enlarged head portion **222** to engage with the window jamb walls and secure the position of the balance shoe **204** within the jamb track. The locked position also enables the pivot bar to be inserted and/or removed from the cam **226** as required or desired. In contrast, FIG. 3C illustrates the cam **226** in an unlocked position such that the keyhole opening **273** is rotated approximately 90° and the ends **230** are retracted at least partially within the enlarged head portion **222**. In the unlocked position, the window balance shoe **204** can slide within the window jamb as the window sash is raised or lowered.

(35) FIGS. 4A and 4B are schematic views of loading forces that act on the window balance system **200**. Referring concurrently to FIGS. 4A and 4B, the window balance system **200** is a block and tackle system that includes the balance shoe **204** that is directly attached to the U-shaped channel **210** of the inverted window balance **202**. The balance shoe **204** is coupled to the U-shaped channel **210** with the connection pocket **238** (shown in FIG. 3A) engaged with the fastener **216**, the engagement projection **244** is engaged with the U-shaped channel **210**, and the protrusion **252** is engaged with the U-shaped channel **210**. As such, three load transfer points are formed, one for each connection. Because the window sash is supported by the balance shoe **204**, via the pivot pins, and the balance spring is supported within the U-shaped channel **210**, the operational loads must be transferred between the U-shaped channel **210** and the balance shoe **204** in order to facilitate hung window operation.

(36) In operation, the weight of the window sash and the movement thereof creates a longitudinal load **274** that acts along the longitudinal axis **241** of the window balance system **200**. This longitudinal load **274** is transferred between the balance shoe **204** and the inverted window balance **202** mostly in shear, and the engagement between the fastener **216** and the connection pocket and the engagement between the protrusion **252** and the U-shaped channel **210** carries the majority of this load. The protrusion **252** generally has a high shear strength and a large surface area upon which the longitudinal load **274** is transferred. As such, the protrusion **252** can increase the load capacity of the balance shoe **204** by 50% or more when compared to known designs (e.g., over that of the hook/connection pocket connection alone). In one example, a rectangular-shaped protrusion **252** may be used with a longer edge positioned substantially orthogonal to the load **274** so that a large surface area is formed to transfer load between.

(37) Additionally, during operation, the balance shoe **204** may be pulled away from the U-shaped channel **210** when installed in a window assembly and creates a pullout load **276** that separates the front face of the balance shoe **204** from the U-shaped channel **210**. This load **276** may be a rotationally induced load (as illustrated), a linear load, or a combination thereof. The engagement between the engagement projection **244** and the U-shaped channel **210** carries the majority of this load **276**. However, in some examples, the protrusion **252** (e.g., via a resilient connection) may also carry the pullout load **276** and resist the front face of the balance shoe **204** pulling away from the U-shaped channel **210**.

(38) In the example, the protrusion **252** is sized and shaped so as to not interfere with the pivot bar as it is being dropped into the balance shoe **204**. By positioning the protrusion **252** in the path of travel of the pivot bar, the longitudinal load generated by the window sash is more in line with protrusion **252** along the longitudinal axis **241** (e.g., both the rotating cam and the protrusion **252** are aligned). This facilitates a stronger and more secure connection. In comparison, at least some known balance shoes that couple to the base wall of the U-shaped channel opposite of the pivot bar channel create a load path that is not aligned, and thereby, generates an inherent undesirable pull out force (e.g., the rotating cam and the protrusion are offset from one another). Furthermore, some known balance system have the U-shaped channel facing outward from the window jamb to help

receive the pivot bar during sash installation. However, this orientation of the U-shaped channel exposes the balance system components (e.g., spring and pulleys) to dirt and debris accumulation. In contrast, the U-shaped channel **210** as described herein is oriented so as to protect the balance system components from dirt and debris accumulation and the configuration of the balance shoe **204** enables this operation.

(39) FIG. 5 is a perspective view of another window balance shoe **300** that may be used with the inverted block and tackle window balance. The window balance shoe **300** has an elongate tail portion **302** that includes a connection pocket **304** and at least one connecting device **306**. The window balance shoe **300** also includes an enlarged head portion **308** that includes a locking device **310** and a cam **312** (e.g., locking system) as described above. The cam **312** includes a keyhole opening **314** that is sized and shaped to receive the pivot bar (not shown) and facilitate the pivotable connection between the window sash and the balance shoe **300**. However, in this example, the elongate tail portion **302** has a length **L 316** that is greater than the previous example described above in FIGS. 2-4B. By lengthening the elongate tail portion **302**, the balance shoe **300** can be coupled to the U-shaped channel so that the enlarged head portion **308** extends further outward from the end of the U-shaped channel. Accordingly, the balance shoe **300** allows a fixed-sized U-shaped channel to be used in a larger window having a greater travel distance by extending the length of the entire window balance system with the selected position of the balance shoe **300**.

(40) Additionally, for larger and heavier window assemblies, one or more protrusions **318** may extend from the elongate tail portion **302**. Accordingly, the connection between the U-shaped channel and the balance shoe **300** can have additional load capacity and resistance to disengagement. For example, in transferring the longitudinal window load and/or the pullout load that increases with the length **316** of the balance shoe **300**. In some examples, the one or more protrusions **318** may eliminate the need for either one or both of the connection pocket **304** and connecting devices **306**. In another example, a screw (not shown) driven through the U-shaped channel and the elongate tail portion **302** may be used in conjunction with the protrusion **318** to secure the balance shoe **300** to the U-shaped channel. In yet other examples, the screw may replace the protrusions **318** entirely.

(41) FIG. 6 is a side view of an exemplary protrusion **400** that may be used with the window balance shoes described above. The protrusion **400** extends from a front surface **402** (e.g., a ramped surface) of a balance shoe as described above. In this example, the protrusion **400** is formed with two symmetrical halves **404** at least partially separated by a gap **406**. Each half **404** has an enlarged bulb **408** and a flexible post **410**. The bulbs **408** are larger in size than that of the opening in the U-shaped channel. As such, when the protrusion **400** is received in the U-shaped channel, each half **404** of the protrusion **400** flexes **F** toward the gap **406** and resiliently-fits within the opening. Once the bulbs **408** pass through the U-shaped channel opening, they can return to their original position and the U-shaped channel is secured about the posts **410**. This enables for the protrusion **400** to carry shear loads as well as pull out forces. In other examples, any other type of connection type may be used that enables the balance shoe to function as described herein. With use of the resilient connection, however, positive feedback is provided to the installer so that correct installation within the U-shaped channel is ensured. Additionally, the resilient connection resists pullout forces so that the front face of the balance shoe does not pull away from the U-shaped channel.

(42) FIG. 7 is a perspective view of another window balance shoe **500**. Similar to the examples described above the window balance shoe **500** has an elongate tail portion **502** that includes a connection pocket **504**, at least one connecting device **506**, and an elongate channel **508**. The window balance shoe **500** also includes an enlarged head portion **510** that houses a locking system (not shown). In this example, however, a protrusion **512** is disposed at least partially within the elongate channel **508** and has a face surface **514** that is substantially square to a tool parting line since the balance shoe **500** is typically a molded component. As such, the face surface **514** is substantially flat and parallel to a back face **516** of the balance shoe **500**. By squaring the face

surface **514** of the protrusion **512** with the tool parting line, flash (e.g., excess material that forms between the core and the cavity halves of the molded part) is reduced or eliminated. As such, the efficiency of the manufacturing process is increased and secondary processes to remove the flash (e.g., by hand trimming, vibratory tumbling, blasting, deflashing, etc.) are reduced or eliminated. (43) Additionally, in this example, the protrusion **512** includes a top wall **518**, a bottom wall **520**, and two sidewalls **522**. The top wall **518** and/or the bottom wall **520** of the protrusion **512** may be curved, while the two sidewalls **522** are substantially parallel to one another. The two sidewalls **522** may also be substantially parallel to sidewalls **524** of the elongate tail portion **502**. In some examples, the sidewalls **522** of the protrusion **512** may be angled relative to one another (e.g., if the protrusion **512** is trapezoidal or triangle in shape). The height of the top wall **518** may correspond to the thickness of the U-shaped channel that the balance shoe **500** couples to so that the protrusion **512** does not interfere with the pivot bar as it is dropped into the locking system. In other examples, the protrusion **512** may extend outward from the U-shaped channel when coupled thereto as required or desired.

(44) FIG. **8** is a flowchart illustrating a method **600** of assembling a block and tackle window balance system. The method **600** includes engaging a balance shoe with a fastener that extends across a U-shaped channel in a first orientation (operation **602**). The U-shaped channel includes a base wall and two opposing sidewalls, and the fastener extends across the two sidewalls. For example, the first orientation may include orienting the balance shoe at an angle relative to the U-shaped channel such that an elongated tail portion is partially inserted into the U-shaped channel and an enlarged head portion is not aligned with the U-shaped channel. The balance shoe is then pivoted into a different second orientation relative to the U-shaped channel (operation **604**). In the second orientation, the elongated tail portion is disposed within the U-shaped channel and the enlarged head portion of the balance shoe extends from the U-shaped channel but is aligned with a longitudinal axis of the window balance system. Substantially simultaneously with pivoting the balance shoe, at least one protrusion of the balance shoe is engaged with at least one corresponding opening that is defined in the base wall of the U-shaped channel (operation **606**). The protrusion extends from a front face of the balance shoe and at least partially within an elongate channel of the balance shoe. The elongate channel is positioned adjacent to the base wall of the U-shaped channel in the second orientation and is configured to allow passage of a pivot bar.

(45) In some examples, the method **600** may further include engaging at least one connecting device of the balance shoe with a sidewall of the U-shaped channel (operation **608**). The connecting device may engage with the U-shaped channel substantially simultaneously with the pivoting of the balance shoe (operation **604**). Additionally, when the connecting device engages with the U-shaped channel, at least a portion of the connecting device may slide through a dimple formed in the sidewall of the U-shaped channel (operation **610**).

(46) The materials utilized in the balance systems described herein may be those typically utilized for window and window component manufacture. Material selection for most of the components may be based on the proposed use of the window. Appropriate materials may be selected for the sash balance systems used on particularly heavy window panels, as well as on windows subject to certain environmental conditions (e.g., moisture, corrosive atmospheres, etc.). Aluminum, steel, stainless steel, zinc, or composite materials can be utilized (e.g., for the shoe locking systems). Bendable and/or moldable plastics may be particularly useful.

(47) Any number of the features of the different examples described herein may be combined into one single example and alternate examples having fewer than or more than all of the features herein described are possible. It is to be understood that terminology employed herein is used for the purpose of describing particular examples only and is not intended to be limiting. It must be noted that, as used in this specification, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

(48) While there have been described herein what are to be considered exemplary and preferred

examples of the present technology, other modifications of the technology will become apparent to those skilled in the art from the teachings herein. The particular methods of manufacture and geometries disclosed herein are exemplary in nature and are not to be considered limiting. It is therefore desired to be secured in the appended claims all such modifications as fall within the spirit and scope of the technology. Accordingly, what is desired to be secured by Letters Patent is the technology as defined and differentiated in the following claims, and all equivalents.

## Claims

1. A balance shoe for a block and tackle window balance system, the balance shoe comprising: an enlarged head portion housing a locking system comprising a rotatable cam having a keyhole opening configured to receive at least a portion of a pivot bar and a locking device configured to releasably engage a jamb track; an elongate tail portion defining a longitudinal axis and comprising a front face and an opposite back face both extending between two opposing sidewalls, the back face defining a connection pocket configured to engage a fastener of a U-shaped channel of the window balance system, and wherein the front face of the elongate tail portion is adjacent to a base wall of the U-shaped channel when the elongate tail portion is coupled therein; at least one protrusion extending from the front face of the elongate tail portion and configured to engage with the base wall of the U-shaped channel; and a connecting device disposed on each sidewall of the two opposing sidewalls and configured to be resiliently secured to a corresponding sidewall of the U-shaped channel, the connecting device comprising a flexible arm extending from the sidewall of the elongate tail portion at a line of flexure along the longitudinal axis and an engagement projection located at a free end of the flexible arm, wherein the line of flexure is disposed between the connection pocket and the at least one protrusion along the longitudinal axis of the elongate tail portion.
2. The balance shoe of claim 1, wherein the engagement projection and the at least one protrusion are aligned relative to the longitudinal axis of the elongate tail portion.
3. The balance shoe of claim 1, wherein the front face defines an elongate channel recessed relative to the two opposing sidewalls, the at least one protrusion disposed within the elongate channel.
4. The balance shoe of claim 3, wherein a top end of the elongate tail portion defines a hook around the connection pocket, the elongate channel extending at least partially around the hook.
5. The balance shoe of claim 1, wherein the at least one protrusion comprises two or more protrusions spaced along the longitudinal axis.
6. The balance shoe of claim 1, wherein the at least one protrusion comprises two symmetrical, flexible halves separated by a gap.
7. The balance shoe of claim 1, wherein the at least one protrusion comprises a top wall, a bottom wall, and two sidewalls, the top wall and the bottom wall are curved and the two sidewalls are parallel to each other.
8. The balance shoe of claim 1, wherein the at least one protrusion comprises a face surface parallel to the back face of the elongate tail portion.
9. The balance shoe of claim 1, wherein the at least one protrusion and the connecting device are configured to be releasable disengaged from the U-shaped channel such that the balance shoe is selectively removable from the U-shaped channel.
10. The balance shoe of claim 1, wherein the flexible arm of the connecting device has a length between the free end and the line of flexure, the length greater than a longitudinal length of the at least one protrusion defined between a top wall and a bottom wall.
11. A block and tackle window balance system comprising: a U-shaped channel comprising a base wall and two opposing sidewalls housing at least partially a block and tackle balance assembly, wherein the U-shaped channel comprises a first end having a fastener extending between the two opposing sidewalls, an aperture defined on each of the two opposing sidewalls, and at least one

opening defined within the base wall, and wherein the aperture and the at least one opening are disposed between the first end and the fastener; and a balance shoe coupled to the first end of the U-shaped channel, wherein the balance shoe comprises: an enlarged head portion extending from the first end of the U-shaped channel; a locking system housed within the enlarged head portion and comprising a rotatable cam having a keyhole opening configured to receive at least a portion of a pivot bar and a locking device configured to releasably engage a jamb track; an elongate tail portion defining a longitudinal axis and comprising a front face and an opposite back face both extending between two opposing sidewalls, the back face defining a connection pocket engaged with the fastener of the U-shaped channel; at least one protrusion extending from the front face of the elongate tail portion and engaged with the at least one opening of the base wall of the U-shaped channel such that the front face of the elongate tail portion is adjacent to the base wall; and a connecting device disposed on each sidewall of the two opposing sidewalls of the elongate tail portion, the connecting device comprising a flexible arm extending from the sidewall of the elongate tail portion at a line of flexure along the longitudinal axis and an engagement projection located at a free end of the flexible arm engaged with the aperture of the sidewall of the U-shaped channel, wherein the line of flexure is disposed between the connection pocket and the at least one protrusion along the longitudinal axis of the elongate tail portion.

12. The block and tackle window balance system of claim 11, wherein the at least one opening and the apertures are axially aligned at the first end of the U-shaped channel.

13. The block and tackle window balance system of claim 11, wherein the base wall of the U-shaped channel comprises a ramped portion at the first end of the U-shaped channel, the at least one opening disposed within the ramped portion.

14. The block and tackle window balance system of claim 13, wherein the at least one protrusion comprises a face surface parallel to the ramped portion of the U-shaped channel.

15. The block and tackle window balance system of claim 11, wherein a dimple is formed on each sidewall of the U-shaped channel proximate the aperture.

16. The block and tackle window balance system of claim 11, wherein the at least one opening comprises two or more openings spaced along the longitudinal axis, and the at least one protrusion comprises two or more corresponding protrusions.

17. The block and tackle window balance system of claim 11, wherein the at least protrusion is spaced apart from the connection pocket on the elongate tail portion such that space therebetween is configured to receive a screw driven through the U-shaped channel into the elongate tail portion.

18. The block and tackle window balance system of claim 11, wherein the at least one protrusion of the balance shoe is configured to at least partially flex and snap-fit engage with the at least one opening of the U-shaped channel.

19. The block and tackle window balance system of claim 11, the at least one protrusion comprises a top wall, a bottom wall, and two sidewalls, the top wall having a height that equals a thickness of the base wall of the U-shaped channel.

20. The block and tackle window balance system of claim 11, wherein the flexible arm of the connecting device has a length between the free end and the line of flexure, the length greater than a longitudinal length of the at least one protrusion defined between a top wall and a bottom wall.

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